Table 1: KM2 estimated-prior diagnostic, specific to the estimator and implementation audited in estimated-prior experiment. Prior panels show the mean estimated/true prior ratio and empirical 5th–95th percentiles across repetitions; performance panels show paired mean changes from the correct supplied prior, in percentage points. The table retains every audited setting and real-data budget for the four directly prior-dependent fitted methods. Simulation cases use 100 repetitions; real-data cases use 20 repeated splits, whose uncertainty is conditional resampling uncertainty rather than independent-population Monte Carlo uncertainty. KM2 findings are not generalized to other prior estimators.

One-sample - simulations / AP

Setting	Method	Prior ratio	Change (pp)	95% interval	n / target
Reference	uPU logistic	2.58	-0.02	[-0.03, +0.00]	100/100
Reference	nnPU logistic	2.58	-0.02	[-0.03, +0.00]	100/100
Reference	AdaPU	2.58	+1.01	[+0.81, +1.20]	100/100
Reference	PU gradient boosting	2.58	-0.26	[-0.47, -0.04]	100/100
Prevalence $\pi = 0.01$	uPU logistic	7.22	+0.01	[+0.00, +0.02]	100/100
Prevalence $\pi = 0.01$	nnPU logistic	7.22	+0.01	[+0.00, +0.02]	100/100
Prevalence $\pi = 0.01$	AdaPU	7.22	+0.04	[-0.09, +0.17]	100/100
Prevalence $\pi = 0.01$	PU gradient boosting	7.22	+0.01	[-0.07, +0.09]	100/100
Prevalence $\pi = 0.05$	uPU logistic	7.51	-0.06	[-0.09, -0.02]	100/100
Prevalence $\pi = 0.05$	nnPU logistic	7.51	-0.06	[-0.10, -0.02]	100/100
Prevalence $\pi = 0.05$	AdaPU	7.51	+0.73	[+0.53, +0.93]	100/100
Prevalence $\pi = 0.05$	PU gradient boosting	7.51	-0.05	[-0.33, +0.24]	100/100
Prevalence $\pi = 0.50$	uPU logistic	1.35	+0.01	[+0.01, +0.02]	100/100
Prevalence $\pi = 0.50$	nnPU logistic	1.35	+0.01	[+0.01, +0.02]	100/100
Prevalence $\pi = 0.50$	AdaPU	1.35	-0.94	[-1.07, -0.80]	100/100
Prevalence $\pi = 0.50$	PU gradient boosting	1.35	-0.02	[-0.05, +0.02]	100/100
High dimension ($p = 500$)	uPU logistic	0.36	+0.27	[+0.25, +0.30]	100/100
High dimension ($p = 500$)	nnPU logistic	0.36	+0.27	[+0.25, +0.30]	100/100
High dimension ($p = 500$)	AdaPU	0.36	-0.71	[-0.88, -0.54]	100/100
High dimension ($p = 500$)	PU gradient boosting	0.36	-0.09	[-0.43, +0.24]	100/100
Curved boundary	uPU logistic	2.70	-0.52	[-0.61, -0.44]	100/100
Curved boundary	nnPU logistic	2.70	-0.53	[-0.61, -0.44]	100/100
Curved boundary	AdaPU	2.70	+1.89	[+1.69, +2.10]	100/100
Curved boundary	PU gradient boosting	2.70	-0.37	[-0.55, -0.20]	100/100
Feature interaction	uPU logistic	2.67	-0.05	[-0.07, -0.02]	100/100
Feature interaction	nnPU logistic	2.67	-0.05	[-0.07, -0.02]	100/100
Feature interaction	AdaPU	2.67	-0.23	[-0.50, +0.04]	100/100
Feature interaction	PU gradient boosting	2.67	-0.79	[-1.07, -0.52]	100/100
Unequal class covariance	uPU logistic	2.35	+0.01	[-0.02, +0.04]	100/100
Unequal class covariance	nnPU logistic	2.35	+0.01	[-0.02, +0.04]	100/100
Unequal class covariance	AdaPU	2.35	-4.61	[-4.95, -4.28]	100/100
Unequal class covariance	PU gradient boosting	2.35	-1.02	[-1.47, -0.56]	100/100

One-sample - simulations / AUROC

Setting	Method	Prior ratio	Change (pp)	95% interval	n / target
Reference	uPU logistic	2.58	-0.01	[-0.02, +0.00]	100/100
Reference	nnPU logistic	2.58	-0.01	[-0.02, +0.00]	100/100
Reference	AdaPU	2.58	+1.69	[+1.57, +1.80]	100/100
Reference	PU gradient boosting	2.58	-0.26	[-0.39, -0.13]	100/100
Prevalence $\pi = 0.01$	uPU logistic	7.22	+0.03	[+0.01, +0.04]	100/100
Prevalence $\pi = 0.01$	nnPU logistic	7.22	+0.03	[+0.01, +0.05]	100/100

One-sample - simulations / AUROC

Setting	Method	Prior ratio	Change (pp)	95% interval	n / target
Prevalence $\pi = 0.01$	AdaPU	7.22	+0.18	[-0.43, +0.80]	100/100
Prevalence $\pi = 0.01$	PU gradient boosting	7.22	+0.07	[-0.68, +0.81]	100/100
Prevalence $\pi = 0.05$	uPU logistic	7.51	-0.07	[-0.11, -0.03]	100/100
Prevalence $\pi = 0.05$	nnPU logistic	7.51	-0.07	[-0.11, -0.03]	100/100
Prevalence $\pi = 0.05$	AdaPU	7.51	+1.45	[+1.21, +1.69]	100/100
Prevalence $\pi = 0.05$	PU gradient boosting	7.51	+0.43	[-0.35, +1.22]	100/100
Prevalence $\pi = 0.50$	uPU logistic	1.35	+0.01	[+0.00, +0.02]	100/100
Prevalence $\pi = 0.50$	nnPU logistic	1.35	+0.01	[+0.00, +0.02]	100/100
Prevalence $\pi = 0.50$	AdaPU	1.35	+0.08	[-0.03, +0.19]	100/100
Prevalence $\pi = 0.50$	PU gradient boosting	1.35	+0.00	[-0.03, +0.02]	100/100
High dimension ($p = 500$)	uPU logistic	0.36	+0.26	[+0.24, +0.28]	100/100
High dimension ($p = 500$)	nnPU logistic	0.36	+0.26	[+0.24, +0.28]	100/100
High dimension ($p = 500$)	AdaPU	0.36	-0.61	[-0.71, -0.52]	100/100
High dimension ($p = 500$)	PU gradient boosting	0.36	-0.16	[-0.40, +0.08]	100/100
Curved boundary	uPU logistic	2.70	-0.04	[-0.06, -0.02]	100/100
Curved boundary	nnPU logistic	2.70	-0.04	[-0.06, -0.02]	100/100
Curved boundary	AdaPU	2.70	+2.49	[+2.35, +2.64]	100/100
Curved boundary	PU gradient boosting	2.70	-0.23	[-0.36, -0.10]	100/100
Feature interaction	uPU logistic	2.67	-0.05	[-0.08, -0.03]	100/100
Feature interaction	nnPU logistic	2.67	-0.05	[-0.08, -0.03]	100/100
Feature interaction	AdaPU	2.67	+2.46	[+2.29, +2.62]	100/100
Feature interaction	PU gradient boosting	2.67	-0.13	[-0.36, +0.10]	100/100
Unequal class covariance	uPU logistic	2.35	+0.00	[-0.01, +0.02]	100/100
Unequal class covariance	nnPU logistic	2.35	+0.00	[-0.01, +0.02]	100/100
Unequal class covariance	AdaPU	2.35	-1.27	[-1.42, -1.11]	100/100
Unequal class covariance	PU gradient boosting	2.35	-0.38	[-0.67, -0.09]	100/100

Two-sample - simulations / AP

Setting	Method	Prior ratio	Change (pp)	95% interval	n / target
Reference	uPU logistic	2.52	+0.00	[-0.02, +0.02]	100/100
Reference	nnPU logistic	2.52	+0.01	[-0.02, +0.03]	100/100
Reference	AdaPU	2.52	+1.00	[+0.79, +1.21]	100/100
Reference	PU gradient boosting	2.52	-0.69	[-0.85, -0.54]	100/100
Prevalence $\pi = 0.01$	uPU logistic	41.67	-0.01	[-0.04, +0.01]	100/100
Prevalence $\pi = 0.01$	nnPU logistic	41.67	-0.01	[-0.04, +0.01]	100/100
Prevalence $\pi = 0.01$	AdaPU	41.67	+0.16	[+0.03, +0.30]	100/100
Prevalence $\pi = 0.01$	PU gradient boosting	41.67	+0.95	[+0.83, +1.07]	100/100
Prevalence $\pi = 0.05$	uPU logistic	8.69	+0.00	[-0.02, +0.01]	100/100
Prevalence $\pi = 0.05$	nnPU logistic	8.69	+0.01	[-0.01, +0.02]	100/100
Prevalence $\pi = 0.05$	AdaPU	8.69	+0.46	[+0.30, +0.62]	100/100
Prevalence $\pi = 0.05$	PU gradient boosting	8.69	+0.63	[+0.52, +0.74]	100/100
Prevalence $\pi = 0.50$	uPU logistic	1.14	+0.04	[-0.04, +0.11]	100/100
Prevalence $\pi = 0.50$	nnPU logistic	1.14	+0.04	[-0.04, +0.11]	100/100
Prevalence $\pi = 0.50$	AdaPU	1.14	-1.66	[-1.89, -1.43]	100/100
Prevalence $\pi = 0.50$	PU gradient boosting	1.14	-0.14	[-0.27, -0.02]	100/100
High dimension ($p = 500$)	uPU logistic	0.07	-0.10	[-0.36, +0.15]	100/100
High dimension ($p = 500$)	nnPU logistic	0.07	-1.69	[-1.76, -1.62]	100/100
High dimension ($p = 500$)	AdaPU	0.07	-1.08	[-1.33, -0.83]	100/100
High dimension ($p = 500$)	PU gradient boosting	0.07	-5.60	[-6.07, -5.14]	100/100
Curved boundary	uPU logistic	2.68	-0.40	[-0.48, -0.31]	100/100

Two-sample - simulations / AP

Setting	Method	Prior ratio	Change (pp)	95% interval	n / target
Curved boundary	nnPU logistic	2.68	-0.40	[-0.49, -0.32]	100/100
Curved boundary	AdaPU	2.68	+1.34	[+1.10, +1.58]	100/100
Curved boundary	PU gradient boosting	2.68	-0.43	[-0.64, -0.21]	100/100
Feature interaction	uPU logistic	2.54	-0.10	[-0.17, -0.02]	100/100
Feature interaction	nnPU logistic	2.54	-0.10	[-0.17, -0.02]	100/100
Feature interaction	AdaPU	2.54	-0.11	[-0.35, +0.14]	100/100
Feature interaction	PU gradient boosting	2.54	+0.05	[-0.10, +0.19]	100/100
Unequal class covariance	uPU logistic	2.34	+0.03	[+0.00, +0.07]	100/100
Unequal class covariance	nnPU logistic	2.34	+0.03	[+0.00, +0.07]	100/100
Unequal class covariance	AdaPU	2.34	-4.55	[-4.85, -4.25]	100/100
Unequal class covariance	PU gradient boosting	2.34	-1.39	[-1.72, -1.07]	100/100

Two-sample - simulations / AUROC

Setting	Method	Prior ratio	Change (pp)	95% interval	n / target
Reference	uPU logistic	2.52	+0.00	[-0.01, +0.02]	100/100
Reference	nnPU logistic	2.52	+0.00	[-0.01, +0.02]	100/100
Reference	AdaPU	2.52	+1.78	[+1.67, +1.89]	100/100
Reference	PU gradient boosting	2.52	-0.18	[-0.29, -0.07]	100/100
Prevalence $\pi = 0.01$	uPU logistic	41.67	+0.01	[-0.01, +0.03]	100/100
Prevalence $\pi = 0.01$	nnPU logistic	41.67	+0.01	[-0.02, +0.03]	100/100
Prevalence $\pi = 0.01$	AdaPU	41.67	+1.58	[+1.34, +1.82]	100/100
Prevalence $\pi = 0.01$	PU gradient boosting	41.67	+5.14	[+4.70, +5.59]	100/100
Prevalence $\pi = 0.05$	uPU logistic	8.69	+0.00	[-0.01, +0.01]	100/100
Prevalence $\pi = 0.05$	nnPU logistic	8.69	+0.00	[-0.02, +0.01]	100/100
Prevalence $\pi = 0.05$	AdaPU	8.69	+1.67	[+1.54, +1.80]	100/100
Prevalence $\pi = 0.05$	PU gradient boosting	8.69	+1.41	[+1.29, +1.53]	100/100
Prevalence $\pi = 0.50$	uPU logistic	1.14	+0.03	[-0.04, +0.11]	100/100
Prevalence $\pi = 0.50$	nnPU logistic	1.14	+0.03	[-0.04, +0.11]	100/100
Prevalence $\pi = 0.50$	AdaPU	1.14	-1.16	[-1.35, -0.98]	100/100
Prevalence $\pi = 0.50$	PU gradient boosting	1.14	-0.10	[-0.20, +0.00]	100/100
High dimension ($p = 500$)	uPU logistic	0.07	-0.06	[-0.38, +0.26]	100/100
High dimension ($p = 500$)	nnPU logistic	0.07	-1.81	[-1.89, -1.74]	100/100
High dimension ($p = 500$)	AdaPU	0.07	-0.94	[-1.08, -0.79]	100/100
High dimension ($p = 500$)	PU gradient boosting	0.07	-3.88	[-4.25, -3.52]	100/100
Curved boundary	uPU logistic	2.68	-0.02	[-0.05, +0.00]	100/100
Curved boundary	nnPU logistic	2.68	-0.03	[-0.05, +0.00]	100/100
Curved boundary	AdaPU	2.68	+2.29	[+2.14, +2.44]	100/100
Curved boundary	PU gradient boosting	2.68	-0.11	[-0.26, +0.03]	100/100
Feature interaction	uPU logistic	2.54	-0.11	[-0.18, -0.05]	100/100
Feature interaction	nnPU logistic	2.54	-0.11	[-0.18, -0.05]	100/100
Feature interaction	AdaPU	2.54	+2.24	[+2.09, +2.38]	100/100
Feature interaction	PU gradient boosting	2.54	+0.10	[+0.04, +0.17]	100/100
Unequal class covariance	uPU logistic	2.34	+0.02	[+0.00, +0.04]	100/100
Unequal class covariance	nnPU logistic	2.34	+0.02	[+0.00, +0.04]	100/100
Unequal class covariance	AdaPU	2.34	-1.31	[-1.44, -1.17]	100/100
Unequal class covariance	PU gradient boosting	2.34	-0.47	[-0.65, -0.28]	100/100

One-sample - real, budget 0.05, AP

Setting	Method	Prior ratio	Change (pp)	95% interval	n / target
Credit Default	uPU logistic	1.60	-1.04	[-1.59, -0.49]	20/20
Credit Default	nnPU logistic	1.60	-1.06	[-1.61, -0.52]	20/20
Credit Default	AdaPU	1.60	-0.42	[-1.41, +0.56]	20/20
Credit Default	PU gradient boosting	1.60	+0.98	[-0.45, +2.40]	20/20
Dry Bean	uPU logistic	0.85	+0.03	[-0.10, +0.16]	20/20
Dry Bean	nnPU logistic	0.85	-0.08	[-0.25, +0.09]	20/20
Dry Bean	AdaPU	0.85	-0.98	[-1.54, -0.43]	20/20
Dry Bean	PU gradient boosting	0.85	-0.49	[-1.15, +0.17]	20/20
HTRU2	uPU logistic	0.77	-0.13	[-0.87, +0.61]	20/20
HTRU2	nnPU logistic	0.77	-0.12	[-0.83, +0.59]	20/20
HTRU2	AdaPU	0.77	-0.76	[-1.35, -0.16]	20/20
HTRU2	PU gradient boosting	0.77	-3.54	[-8.53, +1.45]	20/20
Online Shoppers	uPU logistic	0.70	+0.17	[+0.05, +0.30]	20/20
Online Shoppers	nnPU logistic	0.70	+0.10	[+0.02, +0.19]	20/20
Online Shoppers	AdaPU	0.70	-0.79	[-1.41, -0.17]	20/20
Online Shoppers	PU gradient boosting	0.70	-0.05	[-1.03, +0.93]	20/20
Optical Digits	uPU logistic	0.46	-1.13	[-2.55, +0.30]	20/20
Optical Digits	nnPU logistic	0.46	-3.07	[-4.68, -1.47]	20/20
Optical Digits	AdaPU	0.46	-3.24	[-4.56, -1.92]	20/20
Optical Digits	PU gradient boosting	0.46	-2.22	[-3.80, -0.65]	20/20
Phishing Websites	uPU logistic	0.56	+1.70	[+0.63, +2.78]	20/20
Phishing Websites	nnPU logistic	0.56	-2.58	[-3.93, -1.23]	20/20
Phishing Websites	AdaPU	0.56	-0.57	[-1.50, +0.36]	20/20
Phishing Websites	PU gradient boosting	0.56	-0.82	[-1.92, +0.28]	20/20
Rice	uPU logistic	0.87	+0.14	[+0.00, +0.28]	20/20
Rice	nnPU logistic	0.87	-0.05	[-0.19, +0.09]	20/20
Rice	AdaPU	0.87	-0.43	[-0.85, -0.01]	20/20
Rice	PU gradient boosting	0.87	+0.09	[-0.20, +0.39]	20/20
Spambase	uPU logistic	0.29	-8.24	[-10.04, -6.43]	20/20
Spambase	nnPU logistic	0.29	-9.05	[-10.78, -7.33]	20/20
Spambase	AdaPU	0.29	-3.93	[-7.17, -0.69]	20/20
Spambase	PU gradient boosting	0.29	+3.63	[-0.28, +7.55]	20/20

One-sample - real, budget 0.05, AUROC

Setting	Method	Prior ratio	Change (pp)	95% interval	n / target
Credit Default	uPU logistic	1.60	-0.93	[-1.44, -0.43]	20/20
Credit Default	nnPU logistic	1.60	-0.94	[-1.44, -0.43]	20/20
Credit Default	AdaPU	1.60	-0.04	[-0.67, +0.59]	20/20
Credit Default	PU gradient boosting	1.60	+1.10	[-0.09, +2.30]	20/20
Dry Bean	uPU logistic	0.85	+0.02	[-0.04, +0.09]	20/20
Dry Bean	nnPU logistic	0.85	-0.03	[-0.12, +0.06]	20/20
Dry Bean	AdaPU	0.85	-0.56	[-0.88, -0.24]	20/20
Dry Bean	PU gradient boosting	0.85	-0.29	[-0.73, +0.15]	20/20
HTRU2	uPU logistic	0.77	+0.16	[-0.22, +0.54]	20/20
HTRU2	nnPU logistic	0.77	+0.11	[-0.24, +0.46]	20/20
HTRU2	AdaPU	0.77	-0.50	[-0.99, +0.00]	20/20
HTRU2	PU gradient boosting	0.77	-1.64	[-4.26, +0.98]	20/20
Online Shoppers	uPU logistic	0.70	+0.03	[-0.10, +0.15]	20/20
Online Shoppers	nnPU logistic	0.70	+0.00	[-0.13, +0.13]	20/20
Online Shoppers	AdaPU	0.70	-0.90	[-1.75, -0.06]	20/20

One-sample - real, budget 0.05, AUROC

Setting	Method	Prior ratio	Change (pp)	95% interval	n / target
Online Shoppers	PU gradient boosting	0.70	-0.06	[-1.97, +1.84]	20/20
Optical Digits	uPU logistic	0.46	-1.53	[-2.46, -0.60]	20/20
Optical Digits	nnPU logistic	0.46	-3.70	[-4.91, -2.49]	20/20
Optical Digits	AdaPU	0.46	-2.00	[-2.93, -1.07]	20/20
Optical Digits	PU gradient boosting	0.46	-2.05	[-3.66, -0.44]	20/20
Phishing Websites	uPU logistic	0.56	+1.65	[+0.65, +2.65]	20/20
Phishing Websites	nnPU logistic	0.56	-2.59	[-3.91, -1.28]	20/20
Phishing Websites	AdaPU	0.56	-1.23	[-2.07, -0.40]	20/20
Phishing Websites	PU gradient boosting	0.56	-0.54	[-1.35, +0.27]	20/20
Rice	uPU logistic	0.87	+0.13	[+0.02, +0.25]	20/20
Rice	nnPU logistic	0.87	+0.00	[-0.11, +0.11]	20/20
Rice	AdaPU	0.87	-0.48	[-0.81, -0.14]	20/20
Rice	PU gradient boosting	0.87	+0.03	[-0.19, +0.25]	20/20
Spambase	uPU logistic	0.29	-6.34	[-7.98, -4.70]	20/20
Spambase	nnPU logistic	0.29	-6.76	[-8.03, -5.49]	20/20
Spambase	AdaPU	0.29	-2.00	[-4.29, +0.29]	20/20
Spambase	PU gradient boosting	0.29	+3.15	[-0.44, +6.73]	20/20

One-sample - real, budget 0.20, AP

Setting	Method	Prior ratio	Change (pp)	95% interval	n / target
Credit Default	uPU logistic	1.72	-0.34	[-0.68, -0.01]	20/20
Credit Default	nnPU logistic	1.72	-0.34	[-0.68, -0.01]	20/20
Credit Default	AdaPU	1.72	-1.39	[-1.98, -0.80]	20/20
Credit Default	PU gradient boosting	1.72	-1.33	[-2.10, -0.57]	20/20
Dry Bean	uPU logistic	1.21	-0.52	[-0.65, -0.39]	20/20
Dry Bean	nnPU logistic	1.21	-0.51	[-0.66, -0.36]	20/20
Dry Bean	AdaPU	1.21	+0.69	[+0.37, +1.02]	20/20
Dry Bean	PU gradient boosting	1.21	-1.55	[-2.42, -0.68]	20/20
HTRU2	uPU logistic	0.96	-0.22	[-0.64, +0.19]	20/20
HTRU2	nnPU logistic	0.96	-0.28	[-0.73, +0.18]	20/20
HTRU2	AdaPU	0.96	+0.22	[-0.25, +0.68]	20/20
HTRU2	PU gradient boosting	0.96	+0.00	[-0.54, +0.55]	20/20
Online Shoppers	uPU logistic	2.38	-0.94	[-1.31, -0.57]	20/20
Online Shoppers	nnPU logistic	2.38	-0.66	[-1.09, -0.23]	20/20
Online Shoppers	AdaPU	2.38	+1.87	[+1.06, +2.69]	20/20
Online Shoppers	PU gradient boosting	2.38	+1.36	[+0.42, +2.31]	20/20
Optical Digits	uPU logistic	0.96	-0.16	[-0.46, +0.13]	20/20
Optical Digits	nnPU logistic	0.96	-1.01	[-1.74, -0.28]	20/20
Optical Digits	AdaPU	0.96	-0.46	[-0.96, +0.03]	20/20
Optical Digits	PU gradient boosting	0.96	+0.38	[+0.14, +0.62]	20/20
Phishing Websites	uPU logistic	1.02	-0.08	[-0.27, +0.11]	20/20
Phishing Websites	nnPU logistic	1.02	+0.17	[-0.10, +0.44]	20/20
Phishing Websites	AdaPU	1.02	+0.17	[-0.17, +0.51]	20/20
Phishing Websites	PU gradient boosting	1.02	-0.07	[-0.27, +0.13]	20/20
Rice	uPU logistic	1.00	+0.00	[-0.06, +0.07]	20/20
Rice	nnPU logistic	1.00	-0.01	[-0.08, +0.06]	20/20
Rice	AdaPU	1.00	-0.03	[-0.24, +0.18]	20/20
Rice	PU gradient boosting	1.00	-0.40	[-0.95, +0.16]	20/20
Spambase	uPU logistic	0.89	-0.34	[-0.58, -0.10]	20/20
Spambase	nnPU logistic	0.89	-0.41	[-0.78, -0.05]	20/20

One-sample - real, budget 0.20, AP

Setting	Method	Prior ratio	Change (pp)	95% interval	n / target
Spambase	AdaPU	0.89	+0.12	[-0.28, +0.52]	20/20
Spambase	PU gradient boosting	0.89	+0.77	[+0.32, +1.21]	20/20

One-sample - real, budget 0.20, AUROC

Setting	Method	Prior ratio	Change (pp)	95% interval	n / target
Credit Default	uPU logistic	1.72	-0.50	[-0.68, -0.32]	20/20
Credit Default	nnPU logistic	1.72	-0.50	[-0.68, -0.32]	20/20
Credit Default	AdaPU	1.72	-0.56	[-0.85, -0.28]	20/20
Credit Default	PU gradient boosting	1.72	-0.39	[-1.00, +0.21]	20/20
Dry Bean	uPU logistic	1.21	-0.21	[-0.26, -0.15]	20/20
Dry Bean	nnPU logistic	1.21	-0.20	[-0.27, -0.13]	20/20
Dry Bean	AdaPU	1.21	+0.35	[+0.15, +0.56]	20/20
Dry Bean	PU gradient boosting	1.21	-0.32	[-0.55, -0.08]	20/20
HTRU2	uPU logistic	0.96	-0.02	[-0.11, +0.06]	20/20
HTRU2	nnPU logistic	0.96	-0.13	[-0.24, -0.03]	20/20
HTRU2	AdaPU	0.96	+0.08	[-0.11, +0.26]	20/20
HTRU2	PU gradient boosting	0.96	+0.09	[-0.28, +0.45]	20/20
Online Shoppers	uPU logistic	2.38	-0.57	[-0.88, -0.27]	20/20
Online Shoppers	nnPU logistic	2.38	-0.20	[-0.49, +0.08]	20/20
Online Shoppers	AdaPU	2.38	+1.69	[+0.98, +2.40]	20/20
Online Shoppers	PU gradient boosting	2.38	+0.96	[+0.29, +1.63]	20/20
Optical Digits	uPU logistic	0.96	-0.09	[-0.31, +0.12]	20/20
Optical Digits	nnPU logistic	0.96	-0.77	[-1.21, -0.32]	20/20
Optical Digits	AdaPU	0.96	-0.44	[-0.97, +0.08]	20/20
Optical Digits	PU gradient boosting	0.96	+0.40	[+0.15, +0.65]	20/20
Phishing Websites	uPU logistic	1.02	-0.08	[-0.27, +0.11]	20/20
Phishing Websites	nnPU logistic	1.02	+0.15	[-0.12, +0.43]	20/20
Phishing Websites	AdaPU	1.02	+0.14	[-0.16, +0.44]	20/20
Phishing Websites	PU gradient boosting	1.02	-0.05	[-0.23, +0.12]	20/20
Rice	uPU logistic	1.00	+0.00	[-0.06, +0.05]	20/20
Rice	nnPU logistic	1.00	-0.01	[-0.06, +0.04]	20/20
Rice	AdaPU	1.00	-0.04	[-0.19, +0.11]	20/20
Rice	PU gradient boosting	1.00	-0.15	[-0.35, +0.05]	20/20
Spambase	uPU logistic	0.89	-0.26	[-0.49, -0.02]	20/20
Spambase	nnPU logistic	0.89	-0.41	[-0.67, -0.15]	20/20
Spambase	AdaPU	0.89	+0.12	[-0.23, +0.47]	20/20
Spambase	PU gradient boosting	0.89	+0.69	[+0.32, +1.05]	20/20

Two-sample - real, budget 0.05, AP

Setting	Method	Prior ratio	Change (pp)	95% interval	n / target
Credit Default	uPU logistic	1.67	-1.23	[-2.00, -0.46]	20/20
Credit Default	nnPU logistic	1.67	-1.24	[-2.00, -0.47]	20/20
Credit Default	AdaPU	1.67	-1.14	[-2.10, -0.17]	20/20
Credit Default	PU gradient boosting	1.67	+0.35	[-0.52, +1.21]	20/20
Dry Bean	uPU logistic	1.03	+0.08	[-0.10, +0.25]	20/20
Dry Bean	nnPU logistic	1.03	+0.12	[-0.09, +0.33]	20/20
Dry Bean	AdaPU	1.03	+0.33	[-0.24, +0.90]	20/20
Dry Bean	PU gradient boosting	1.03	-0.12	[-0.50, +0.25]	20/20

Two-sample - real, budget 0.05, AP

Setting	Method	Prior ratio	Change (pp)	95% interval	n / target
HTRU2	uPU logistic	0.73	-0.69	[-1.17, -0.22]	20/20
HTRU2	nnPU logistic	0.73	-0.77	[-1.18, -0.36]	20/20
HTRU2	AdaPU	0.73	-0.87	[-1.95, +0.21]	20/20
HTRU2	PU gradient boosting	0.73	-0.06	[-0.47, +0.36]	20/20
Online Shoppers	uPU logistic	0.67	+0.15	[+0.00, +0.30]	20/20
Online Shoppers	nnPU logistic	0.67	+0.07	[-0.11, +0.26]	20/20
Online Shoppers	AdaPU	0.67	-0.57	[-1.00, -0.14]	20/20
Online Shoppers	PU gradient boosting	0.67	-0.42	[-1.25, +0.41]	20/20
Optical Digits	uPU logistic	0.40	-1.12	[-2.51, +0.28]	20/20
Optical Digits	nnPU logistic	0.40	-2.67	[-4.10, -1.23]	20/20
Optical Digits	AdaPU	0.40	-3.83	[-4.66, -3.01]	20/20
Optical Digits	PU gradient boosting	0.40	+2.63	[+0.79, +4.48]	20/20
Phishing Websites	uPU logistic	0.48	+1.43	[-0.06, +2.93]	20/20
Phishing Websites	nnPU logistic	0.48	-2.16	[-3.53, -0.78]	20/20
Phishing Websites	AdaPU	0.48	-1.70	[-2.56, -0.83]	20/20
Phishing Websites	PU gradient boosting	0.48	-0.28	[-0.95, +0.38]	20/20
Rice	uPU logistic	0.84	+0.28	[-0.11, +0.68]	20/20
Rice	nnPU logistic	0.84	-0.40	[-0.73, -0.08]	20/20
Rice	AdaPU	0.84	+0.26	[-0.19, +0.72]	20/20
Rice	PU gradient boosting	0.84	-0.12	[-0.50, +0.26]	20/20
Spambase	uPU logistic	0.19	-6.34	[-7.69, -4.99]	20/20
Spambase	nnPU logistic	0.19	-7.33	[-8.42, -6.23]	20/20
Spambase	AdaPU	0.19	-0.57	[-2.20, +1.06]	20/20
Spambase	PU gradient boosting	0.19	+0.14	[-2.49, +2.76]	20/20

Two-sample - real, budget 0.05, AUROC

Setting	Method	Prior ratio	Change (pp)	95% interval	n / target
Credit Default	uPU logistic	1.67	-0.84	[-1.38, -0.31]	20/20
Credit Default	nnPU logistic	1.67	-0.83	[-1.37, -0.29]	20/20
Credit Default	AdaPU	1.67	-0.16	[-0.69, +0.36]	20/20
Credit Default	PU gradient boosting	1.67	+0.22	[-0.36, +0.80]	20/20
Dry Bean	uPU logistic	1.03	+0.06	[-0.04, +0.17]	20/20
Dry Bean	nnPU logistic	1.03	+0.08	[-0.04, +0.20]	20/20
Dry Bean	AdaPU	1.03	+0.25	[-0.09, +0.59]	20/20
Dry Bean	PU gradient boosting	1.03	-0.10	[-0.23, +0.03]	20/20
HTRU2	uPU logistic	0.73	+0.03	[-0.48, +0.54]	20/20
HTRU2	nnPU logistic	0.73	+0.00	[-0.34, +0.35]	20/20
HTRU2	AdaPU	0.73	-0.41	[-1.01, +0.19]	20/20
HTRU2	PU gradient boosting	0.73	+0.17	[-0.03, +0.37]	20/20
Online Shoppers	uPU logistic	0.67	+0.16	[+0.00, +0.32]	20/20
Online Shoppers	nnPU logistic	0.67	+0.12	[-0.11, +0.34]	20/20
Online Shoppers	AdaPU	0.67	-0.51	[-0.98, -0.05]	20/20
Online Shoppers	PU gradient boosting	0.67	-0.59	[-1.33, +0.14]	20/20
Optical Digits	uPU logistic	0.40	-1.43	[-2.21, -0.65]	20/20
Optical Digits	nnPU logistic	0.40	-3.52	[-4.74, -2.30]	20/20
Optical Digits	AdaPU	0.40	-2.86	[-3.61, -2.11]	20/20
Optical Digits	PU gradient boosting	0.40	+1.88	[+0.64, +3.12]	20/20
Phishing Websites	uPU logistic	0.48	+1.08	[-0.36, +2.52]	20/20
Phishing Websites	nnPU logistic	0.48	-2.21	[-3.59, -0.83]	20/20
Phishing Websites	AdaPU	0.48	-2.16	[-2.94, -1.37]	20/20

Two-sample - real, budget 0.05, AUROC

Setting	Method	Prior ratio	Change (pp)	95% interval	n / target
Phishing Websites	PU gradient boosting	0.48	-0.02	[-0.43, +0.40]	20/20
Rice	uPU logistic	0.84	+0.26	[-0.08, +0.61]	20/20
Rice	nnPU logistic	0.84	-0.31	[-0.57, -0.04]	20/20
Rice	AdaPU	0.84	-0.07	[-0.29, +0.15]	20/20
Rice	PU gradient boosting	0.84	-0.05	[-0.34, +0.25]	20/20
Spambase	uPU logistic	0.19	-4.47	[-5.59, -3.36]	20/20
Spambase	nnPU logistic	0.19	-5.67	[-6.52, -4.82]	20/20
Spambase	AdaPU	0.19	+0.24	[-0.80, +1.29]	20/20
Spambase	PU gradient boosting	0.19	+0.37	[-1.73, +2.48]	20/20

Two-sample - real, budget 0.20, AP

Setting	Method	Prior ratio	Change (pp)	95% interval	n / target
Credit Default	uPU logistic	1.75	-0.35	[-0.68, -0.02]	20/20
Credit Default	nnPU logistic	1.75	-0.35	[-0.68, -0.02]	20/20
Credit Default	AdaPU	1.75	-1.12	[-1.72, -0.53]	20/20
Credit Default	PU gradient boosting	1.75	-0.04	[-0.42, +0.33]	20/20
Dry Bean	uPU logistic	1.29	-0.56	[-0.69, -0.43]	20/20
Dry Bean	nnPU logistic	1.29	-0.54	[-0.71, -0.37]	20/20
Dry Bean	AdaPU	1.29	+1.16	[+0.77, +1.54]	20/20
Dry Bean	PU gradient boosting	1.29	-0.59	[-0.89, -0.29]	20/20
HTRU2	uPU logistic	0.85	-0.38	[-0.78, +0.02]	20/20
HTRU2	nnPU logistic	0.85	-0.51	[-0.89, -0.12]	20/20
HTRU2	AdaPU	0.85	+0.07	[-0.47, +0.61]	20/20
HTRU2	PU gradient boosting	0.85	+0.44	[-0.29, +1.17]	20/20
Online Shoppers	uPU logistic	1.99	-0.61	[-0.90, -0.32]	20/20
Online Shoppers	nnPU logistic	1.99	-0.37	[-0.68, -0.07]	20/20
Online Shoppers	AdaPU	1.99	+1.08	[+0.46, +1.71]	20/20
Online Shoppers	PU gradient boosting	1.99	-0.06	[-0.63, +0.52]	20/20
Optical Digits	uPU logistic	0.89	+0.22	[-0.09, +0.52]	20/20
Optical Digits	nnPU logistic	0.89	-0.33	[-1.02, +0.36]	20/20
Optical Digits	AdaPU	0.89	-1.00	[-1.69, -0.30]	20/20
Optical Digits	PU gradient boosting	0.89	+0.93	[+0.17, +1.68]	20/20
Phishing Websites	uPU logistic	0.99	+0.03	[-0.11, +0.16]	20/20
Phishing Websites	nnPU logistic	0.99	-0.35	[-1.02, +0.32]	20/20
Phishing Websites	AdaPU	0.99	-0.18	[-0.75, +0.38]	20/20
Phishing Websites	PU gradient boosting	0.99	+0.02	[-0.09, +0.12]	20/20
Rice	uPU logistic	0.99	+0.06	[+0.00, +0.12]	20/20
Rice	nnPU logistic	0.99	-0.01	[-0.07, +0.06]	20/20
Rice	AdaPU	0.99	+0.05	[-0.04, +0.14]	20/20
Rice	PU gradient boosting	0.99	+0.06	[-0.27, +0.38]	20/20
Spambase	uPU logistic	0.82	-0.47	[-1.02, +0.09]	20/20
Spambase	nnPU logistic	0.82	-0.85	[-1.41, -0.30]	20/20
Spambase	AdaPU	0.82	-0.48	[-1.02, +0.06]	20/20
Spambase	PU gradient boosting	0.82	+0.46	[-0.16, +1.09]	20/20

Two-sample - real, budget 0.20, AUROC

Setting	Method	Prior ratio	Change (pp)	95% interval	n / target
Credit Default	uPU logistic	1.75	-0.41	[-0.58, -0.25]	20/20

Two-sample - real, budget 0.20, AUROC

Setting	Method	Prior ratio	Change (pp)	95% interval	n / target
Credit Default	nnPU logistic	1.75	-0.41	[-0.58, -0.25]	20/20
Credit Default	AdaPU	1.75	-0.59	[-0.90, -0.28]	20/20
Credit Default	PU gradient boosting	1.75	+0.06	[-0.17, +0.28]	20/20
Dry Bean	uPU logistic	1.29	-0.20	[-0.27, -0.14]	20/20
Dry Bean	nnPU logistic	1.29	-0.19	[-0.28, -0.11]	20/20
Dry Bean	AdaPU	1.29	+0.65	[+0.45, +0.85]	20/20
Dry Bean	PU gradient boosting	1.29	-0.18	[-0.25, -0.11]	20/20
HTRU2	uPU logistic	0.85	+0.06	[-0.10, +0.22]	20/20
HTRU2	nnPU logistic	0.85	+0.04	[-0.12, +0.20]	20/20
HTRU2	AdaPU	0.85	+0.00	[-0.28, +0.28]	20/20
HTRU2	PU gradient boosting	0.85	+0.44	[-0.05, +0.92]	20/20
Online Shoppers	uPU logistic	1.99	-0.48	[-0.69, -0.27]	20/20
Online Shoppers	nnPU logistic	1.99	-0.31	[-0.51, -0.10]	20/20
Online Shoppers	AdaPU	1.99	+0.62	[+0.13, +1.11]	20/20
Online Shoppers	PU gradient boosting	1.99	+0.02	[-0.28, +0.33]	20/20
Optical Digits	uPU logistic	0.89	+0.10	[-0.08, +0.29]	20/20
Optical Digits	nnPU logistic	0.89	-0.99	[-1.62, -0.36]	20/20
Optical Digits	AdaPU	0.89	-0.76	[-1.38, -0.13]	20/20
Optical Digits	PU gradient boosting	0.89	+0.70	[+0.12, +1.28]	20/20
Phishing Websites	uPU logistic	0.99	+0.00	[-0.13, +0.14]	20/20
Phishing Websites	nnPU logistic	0.99	-0.36	[-1.04, +0.32]	20/20
Phishing Websites	AdaPU	0.99	-0.11	[-0.48, +0.26]	20/20
Phishing Websites	PU gradient boosting	0.99	+0.00	[-0.05, +0.05]	20/20
Rice	uPU logistic	0.99	+0.05	[+0.00, +0.10]	20/20
Rice	nnPU logistic	0.99	-0.01	[-0.06, +0.04]	20/20
Rice	AdaPU	0.99	+0.01	[-0.05, +0.06]	20/20
Rice	PU gradient boosting	0.99	+0.02	[-0.07, +0.10]	20/20
Spambase	uPU logistic	0.82	-0.32	[-0.80, +0.17]	20/20
Spambase	nnPU logistic	0.82	-0.68	[-1.04, -0.31]	20/20
Spambase	AdaPU	0.82	-0.58	[-1.13, -0.03]	20/20
Spambase	PU gradient boosting	0.82	+0.42	[-0.06, +0.90]	20/20

Note. KM2 only. Prior ratio is mean estimated/true prevalence; n refers to matched performance pairs. Simulation intervals quantify within-setting Monte Carlo error; real-data intervals are conditional repeated-split uncertainty and not a population sampling guarantee. All available cases are retained; no failed fits are imputed.